

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)			Substitution: $m = \frac{10\,000}{10}$ (1) = 1 000 [kg] (1)	1	1		2	2	
	(b)	(i)		5 000 – 2 000 (1) = 3 000 [N] (1)		2		2	1	
		(ii)		Substitution: $\frac{3\,000 \text{ ecf}}{1000 \text{ ecf}}$ (1) = 3 (1) m/s ² (1)	1 1	1		3	2	
		(iii)	I	Resultant force decreases (1) because {air resistance / drag / friction / resistive forces} increases (1)		2		2		
			II	Decreases		1		1		
	(c)	(i)		{Energy losses / energy transfers / work done / heat losses} due to {air resistance / drag / friction / resistive forces} Accept no energy had been lost due to friction [at the top of the hill]		1		1		
		(ii)		Substitution: $\frac{72\,000}{15}$ (1) = 4 800 [N] (1)	1	1		2	2	
				Question 9 total	4	9	0	13	8	0